

§12. #5.

We want to show that $\|T_h\|_h^2 \leq 3(\|f\|_h^2 + \|f\|_{L^2(0,1)}^2)$. By Hint, we integrate that

$$T_h(X_j) = \frac{1}{h^2} (R_+(X_j+h) + R_-(X_j-h)) = \frac{1}{h^2} \left(\int_{X_j}^{X_j+h} (u''(t) - u'(X_j)) \frac{(X_j+h-t)^2}{2} dt - \int_{X_j-h}^{X_j} (u''(t) - u'(X_j)) \frac{(X_j-h-t)^2}{2} dt \right)$$

$\therefore$ We get $T_h(X_j) = -u'(X_j) - \frac{1}{h^2} \left[\int_{X_j-h}^{X_j} u''(t) (X_j-h-t)^2 dt - \int_{X_j}^{X_j+h} u''(t) (X_j+h-t)^2 dt \right]$

Let $f(t) = -u''(t)$, so $T_h(X_j) = f(X_j) + \frac{1}{h^2} \left[\int_{X_j-h}^{X_j} f(t) (X_j-h-t)^2 dt - \int_{X_j}^{X_j+h} f(t) (X_j+h-t)^2 dt \right]$

Set $a = f(X_j)$, $b = \frac{1}{h^2} \int_{X_j-h}^{X_j} f(t) (X_j-h-t)^2 dt$, $c = \frac{1}{h^2} \int_{X_j}^{X_j+h} f(t) (X_j+h-t)^2 dt$

By $(a+b-c)^2 \leq 3(a^2+b^2+c^2) \Rightarrow [T_h(X_j)]^2 \leq 3(f(X_j)^2 + b^2 + c^2) \Rightarrow \|T_h\|_h^2 \leq 3(\|f\|_h^2 + h \sum_j b^2 + h \sum_j c^2)$

Let $I_1 = [X_j-h, X_j]$, $\therefore b^2 = \frac{1}{h^4} \left(\int_{I_1} f(t) (X_j-h-t)^2 dt \right)^2 \leq \frac{1}{h^4} \left(\int_{I_1} |f(t)|^2 dt \right) \left(\int_{I_1} (X_j-h-t)^4 dt \right)$

and let $v = X_j-h-t$, $\therefore b^2 \leq \frac{1}{h^4} \left(\int_{I_1} |f(t)|^2 dt \right) \cdot \left(\int_0^h v^4 dv \right) \leq \frac{1}{h^4} \cdot \frac{h^5}{5} \left(\int_{I_1} |f(t)|^2 dt \right)$

so $b^2 \leq \frac{h}{5} \left(\int_{I_1} |f(t)|^2 dt \right)$, similarly for $c^2 \leq \frac{h}{5} \left(\int_{I_2} |f(t)|^2 dt \right)$, which $I_2 = [X_j, X_j+h]$

Hence $h \sum_j b^2 \leq \frac{h^2}{5} \sum_j \left(\int_{I_1} |f(t)|^2 dt \right) = \frac{h^2}{5} \int_0^1 |f(t)|^2 dt = \frac{h^2}{5} \|f\|_{L^2(0,1)}^2$

and $h \sum_j c^2 \leq \frac{h^2}{5} \sum_j \left(\int_{I_2} |f(t)|^2 dt \right) = \frac{h^2}{5} \int_0^1 |f(t)|^2 dt = \frac{h^2}{5} \|f\|_{L^2(0,1)}^2$

$\therefore \|T_h\|_h^2 \leq 3(\|f\|_h^2 + \frac{2h^2}{5} \|f\|_{L^2(0,1)}^2) \leq 3(\|f\|_h^2 + \|f\|_{L^2(0,1)}^2)$, which $0 < h \leq 1$ #

§12. #7. Let $g=1$, by def of (12.25), $T_h g(X_j) = \sum_{k=1}^{n-1} g(X_k) G^k(X_j, X_k)$

From the exercise in HW7, $G^k(X_j) = h G(X_j, X_k)$, $\therefore T_h g(X_j) = h \left[\sum_{k=1}^j X_k (1-X_j) + \sum_{k=j+1}^{n-1} X_j (1-X_k) \right]$

$T_h g(X_j) = h \left[(1-X_j) \sum_{k=1}^j X_k + X_j \sum_{k=j+1}^{n-1} (1-X_k) \right]$

Consider $\sum_{k=1}^j X_k = \sum_{k=1}^j kh = h \cdot \frac{j(j+1)}{2}$ and $\sum_{k=j+1}^{n-1} (1-X_k) = \sum_{k=j+1}^{n-1} 1 - \sum_{k=j+1}^{n-1} X_k = (n-1-j) - h \sum_{k=j+1}^{n-1} k$
 $= (n-1-j) - h \left(\frac{(n-1)h}{2} - \frac{j(j+1)}{2} \right)$

$\therefore T_h g(X_j) = h \left[(1-X_j) \left(h \frac{j(j+1)}{2} \right) + X_j \left[(n-1-j) - h \left(\frac{(n-1)h}{2} - \frac{j(j+1)}{2} \right) \right] \right]$
 $= h \left[(1-jh) \left(h \frac{j(j+1)}{2} \right) + jh \left[(n-1-j) - \frac{h}{2} (n-1)h + \frac{h}{2} j(j+1) \right] \right]$
 $= \frac{h^2}{2} (1-jh)(j+1)j + (n-1-j)jh - \frac{1}{2} jh^2 (n-1)h + \frac{1}{2} jh^2 j(j+1)$
 $= \frac{j(n-j)}{2h^2} \xrightarrow{\downarrow} \frac{1}{2} \cdot \frac{j}{h} \left(1 - \frac{j}{h} \right) = \frac{1}{2} X_j (1-X_j) \quad \#$

$X_j = jh = j \cdot \frac{1}{h}$

§12. #8. It's a special case of Young's inequality. $\Rightarrow ab \leq \varepsilon a^2 + \frac{1}{4\varepsilon} b^2 \quad \forall a, b \in \mathbb{R}, \forall \varepsilon > 0$.

Consider $(\sqrt{\varepsilon}a - \frac{1}{\sqrt{\varepsilon}}b)^2 \geq 0 \Rightarrow \underbrace{\varepsilon a^2}_{\geq 0} - ab + \underbrace{\frac{1}{4\varepsilon} b^2}_{\geq 0} \geq 0 \Rightarrow \varepsilon a^2 + \frac{1}{4\varepsilon} b^2 \geq ab \quad \#$

§12. #9. Show that $\|V_h\|_h \leq \|V_h\|_{h,\infty} \quad \forall h \in V_h$.

sol: $\|V_h\|_h = \left(h \sum_{j=0}^n C_j V_j^2\right)^{\frac{1}{2}}, \quad \|V_h\|_{h,\infty} = \max_{0 \leq j \leq n} |V_h(x_j)|$. Let $M = \|V_h\|_{h,\infty}$, we have that for each V_j , $|V_j| \leq M$, $\therefore V_j^2 \leq M^2$, $\|V_h\|_h^2 = h \sum_{j=0}^n C_j V_j^2 \leq h \sum_{j=0}^n C_j M^2 \leq h M^2 (C_0 + C_1 + \dots + C_n)$
 $\therefore$ with $C_0 = C_n = \frac{1}{2}$ and $C_j = 1, j=1, 2, \dots, n-1$, $\|V_h\|_h^2 \leq h \cdot M^2 (\frac{1}{2} + 1 + \dots + \frac{1}{2}) = h M^2 \leq M^2 \quad (\because 0 < h \leq 1)$
 $\therefore \|V_h\|_h \leq M = \max_{0 \leq j \leq n} |V_h(x_j)| = \|V_h\|_{h,\infty} \quad \#$

§12. #11. The operator $L_h u(x) = -u^{(iv)}(x)$ using centered finite difference

consider the function u_h , $(L_h u_h)(x_j) = -\frac{u_{j+1} - 2u_j + u_{j-1}}{h^2}, \quad j=1, 2, \dots, n-1$
 $\Rightarrow \therefore$ We denote there $(L_h u)_j = -\frac{u_{j+1} - 2u_j + u_{j-1}}{h^2} = \frac{-u_{j+1} + 2u_j - u_{j-1}}{h^2}$

Compute $L_h(L_h u)_j = -\frac{(L_h u)_{j+1} - 2(L_h u)_j + (L_h u)_{j-1}}{h^2} = \frac{[-(L_h u)_{j+1} + 2(L_h u)_j - (L_h u)_{j-1}]}{h^2}$
 $= \frac{[u_{j+2} - 2u_{j+1} + u_j - 2u_{j+1} + 4u_j - 2u_{j-1} + u_j - 2u_{j-1} + u_{j-2}]}{h^4}$
 $= \frac{u_{j+2} - 4u_{j+1} + 6u_j - 4u_{j-1} + u_{j-2}}{h^4}$

Since $L_h u(x) = -u^{(iv)}(x) \Rightarrow L_h u(x) = -\frac{(u_{j+2} - 4u_{j+1} + 6u_j - 4u_{j-1} + u_{j-2})}{h^4} \quad \#$